Confidence-Aware Personalized Adaptive Audio Controller for AI-Generated VR Meditation

Project Proposal Report

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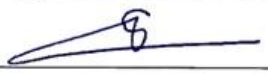
August 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Virtual reality (VR) meditation offers immersive stress-reduction experiences, but existing systems often rely on static or simple adaptive audio cues, limiting personalization and robustness. This proposal details an innovative component for a larger AI-based VR meditation system: a **confidence-aware personalized audio controller**. Our controller will generate a base personalized meditation soundtrack via Google's Lyria RealTime music generator, guided by user preferences collected through a mobile app. It will receive the user's real-time stress level (Component B) and a physiological signal confidence score (Component A) to adapt audio parameters (intensity, density, brightness, etc.) on the fly. Initially, rule-based constraints ensure safe, gradual audio changes; thereafter, reinforcement learning (RL) will optimize personalization based on user feedback. Crucially, the controller manages **both** the generative meditation sound layer and ambient VR audio mix, balancing novelty and familiarity to prevent habituation. For example, when confidence in physiological data is low, adaptation will be conservative to avoid disturbing the user, whereas high-confidence inputs can safely trigger more dynamic audio shifts. A novelty-management strategy will inject controlled variations (dishabituation) to sustain attention without breaking calmness[1][2]. The final output is a seamless, therapeutic audio stream integrated with the VR environment. We will implement the system in Unity (or Unreal) using middleware (Wwise/FM?) and evaluate it through user studies against static and non-personalized baselines. Metrics include relaxation improvement, immersion, stability, and user satisfaction[3][4]. Ethical considerations (data privacy, informed consent, safe design) are emphasized. This novel hybrid controller advances VR meditation by jointly adapting generative meditation music and ambient audio with stress and confidence awareness, a gap in current systems[3][5].

ACKNOWLEDGEMENT

I thank my project supervisor and the VR research group at SLIIT for guidance. I also acknowledge the open-source AI research community (e.g., Google's DeepMind) for tools like Lyria RealTime. My appreciation goes to study participants who will provide feedback for validating this system.

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1. INTRODUCTION

1.1 Background and Context

High stress and anxiety are pervasive global issues, affecting mental health and productivity[6][5]. Mindfulness meditation is an evidence-based stress reduction technique[7]. Immersive VR meditation environments promise greater engagement than traditional methods because users are fully surrounded by multi-sensory cues[3][8]. VR mindfulness applications (e.g., TRIPP) have shown feasibility and short-term effectiveness in reducing stress[9][5]. In particular, VR's immersive nature can enhance focus and presence, helping users remain in the present moment[8][4]. Integrating biofeedback (e.g., breathing, heart rate) further improves outcomes: for example, biofeedback-driven VR relaxation produced greater stress reduction and presence than VR alone[3].

However, most current VR meditation and relaxation systems use **static or canned audio** (fixed guided voice, looped nature sounds) or very simple adaptive rules (e.g., volume up if stress high). They rarely generate truly personalized music, adapt continuously to user state, or consider the *confidence* of physiological measurements. A systematic review found that existing VR relaxation apps largely use unchanging natural environments and fixed sounds[5]. The lack of real-time adaptivity and personalization means these systems may not fully engage users or account for unreliable sensor data. Repetitive audio also leads to habituation (diminished attention and efficacy)[2][1], while abrupt novelty can break calmness.

Proposed solution: We introduce a *Confidence-Aware Personalized Adaptive Audio Controller* (our component) for an AI-driven VR meditation system. It will generate a base customized ambient music layer using **Google Lyria RealTime** (AI music generation), conditioned on user preferences (e.g., favored instruments, moods). It will then dynamically adjust audio parameters (intensity, density, brightness, spatial mix) in response to live stress-level input and a **confidence score** indicating physiological signal reliability. Adaptation starts as conservative rule-based changes (to ensure safety under uncertainty) and evolves into a learned RL policy for fine-grained personalization. Importantly, the controller jointly manages both the meditation music and the ambient VR soundscape, harmonizing them to maintain immersion while responding to user state. This approach is novel in being *confidence-aware* (sensing when data is uncertain and throttling changes) and in balancing familiarity vs novelty to sustain attention without overwhelming the user[2][1].

This project aligns with SDG 3 (Good Health and Well-Being) by advancing stress management tools, and SDG 9 (Industry, Innovation and Infrastructure) through VR and AI innovation. It intersects research clusters of affective computing, multimedia systems, and human-computer interaction.

1.2 Literature Review

VR Meditation Systems: VR relaxation and mindfulness have grown as research fields. Reviews show that VR nature scenes and guided experiences are feasible and effective for stress reduction[5][3]. For example, Chittaro et al. demonstrated a VR breathing-relaxation environment where users' breathing, heart rate, and skin conductance controlled narrative elements; real-time multi-signal biofeedback produced significantly greater relaxation and presence than a sham condition[3]. TRIPP Inc. is a prominent VR meditation product, delivering millions of sessions and demonstrating mood benefits in preliminary trials[10][11]. TRIPP also highlighted the importance of physiological data: "we captured physiological data like heart rate and EEG to see if experiences achieved effect"[11]. These systems use VR to enhance immersion (e.g., particle visuals for breath) and incorporate biofeedback, but generally do not dynamically generate personalized audio content.

Adaptive Audio and Generative Music: Adaptive music in games and VR has been shown to increase immersion. Melodrive's experiment found that "deep adaptive music" (AI-generated in real-time based on scene context) boosted VR immersion by ~30% compared to static music[12]. This suggests dynamic audio fitting the user's state can significantly impact experience. In games, procedural audio and generative soundtracks (e.g., via deep learning) are an active area [8], but often context-driven rather than personalized to the individual's preferences and physiology. Google's DeepMind recently released **Lyria RealTime**, a real-time music generation model allowing moment-by-moment control of parameters (key, tempo, density, brightness) via text prompts[13][14]. It can continuously "jam" and adapt music to user input[14]. However, using Lyria in a closed-loop bioadaptive VR setting is unexplored.

Personalization via Preferences: There is growing work on personalization in VR meditation. For instance, ASafePlace (Lu et al., 2026) lets users co-create a "safe place" VR scene via art and uses an LLM to generate personalized spoken relaxation scripts referring to the user's memory cues[15][16]. This underscores that tailoring content (visuals, audio cues) to user's personal experiences enhances relevance and relaxation. Similarly, music therapy research shows that personalized music (e.g. based on preferred genres or memories) improves engagement and mood. Yet, most systems tailor only visuals or guided instructions, not ambient/generated music.

Biofeedback and Physiological Confidence: VR meditation systems often use physiological signals (heart rate, respiration, EEG) to modulate the environment. Studies like Blum et al. (2019) found that embedding HRV biofeedback in a nature VR environment (versus a standard biofeedback app) resulted in **reduced mind-wandering and increased present-moment focus**[4]. This aligns with theories that VR biofeedback can improve interoception and calmness. However, sensor noise and artifacts (e.g., poor wearables signal) can make these signals unreliable. The concept of a "confidence score" for physiological inputs acknowledges this: when signals are weak or noisy, the system should be conservative to avoid erratic adaptation. In machine learning, confidence measures have been used in classification, but here we apply it to real-time control. No existing VR meditation works, to our knowledge, explicitly incorporate such a confidence-weighted adaptation layer.

Reinforcement Learning for Personalization: RL has been applied to adaptive music (e.g., learning user-preference rewards) and interactive media control[17][18]. For example, Jaques et al. (2017) used RL to fine-tune music generation to maximize user-specified rewards, and later

works propose actor-critic models for real-time interactive music[18]. In VR or game contexts, RL controllers have been used to adjust haptic or difficulty settings based on user feedback[19][20]. In our component, an RL agent will learn from aggregated user responses (explicit feedback or implicit via stress measures) to optimize audio adaptation policies over time. Initially, we will rely on expert-designed rules to ensure safe transitions; then an RL layer will gradually take over personalized decisions, subject to safety constraints (a kind of "safe RL").

Habituation and Novelty: Psychological studies show that people habituate to constant stimuli[2][1]. In meditation, repeating a mantra or sound can lose its effectiveness over time as attention wanes[2]. Conversely, too much novelty can be distracting. Developmental psychology notes that the brain tends to attend to novelty after habituation occurs[1]. In musical context, if a track loops indefinitely, users may stop noticing it. Thus, our design must balance **familiarity** (to keep calmness) with occasional **micro-variations or "dishabituation"** events (to recapture attention)[2][1]. Some adaptive music systems implement controlled variability to avoid monotony, but this is rarely studied in therapeutic audio. We will draw on these insights to include a novelty management strategy: when a segment has played long, slightly shift tempo/timbre or re-generate a compatible variant to refresh the auditory scene without breaking flow.

Audio Parameter Control: Generative models like Lyria allow fine control over musical attributes. The DeepMind Lyria docs describe real-time control of *key, tempo, density, brightness*, blending genres or prompts[13]. These align with our needed dimensions: *intensity/power, rhythmic density, spectral brightness, calmness (e.g. ambient reverb) and* fade/pulse. *Spatial blending (ambient vs meditation sounds) will be managed separately. There is research on audio mixing in VR (binaural ambisonics, spatial audio for immersion), but the novel part is algorithmically adjusting mix ratios based on stress state. Overall, while several pieces of this technology exist (VR, biofeedback, AI music, RL), no prior work has combined* all* these elements into a confidence-aware adaptive audio controller for VR meditation.

1.3 Research Gap

Current VR meditation apps fail to **jointly personalize generated meditation music and ambient audio** based on physiological feedback and its reliability. Most systems use fixed tracks or simple loops, with at best rule-based adjustments to a single parameter (e.g. volume) using one bio-signal[5][3]. They overlook the uncertainty inherent in sensing (no confidence weighting) and do not leverage modern generative AI to create custom soundscapes. There is a lack of a framework that simultaneously (a) uses an AI music generator conditioned on user profile, (b) adapts multiple audio parameters in real time using stress and signal confidence, and (c) mixes meditation music with VR ambient sounds in a balanced way. Moreover, no system addresses the *habituation–novelty trade-off* in an audio therapeutic context. Thus, we propose to fill this gap by developing a **confidence-aware hybrid controller** that uses a user's preference profile, their current stress level, and a reliability score of that stress signal to modulate a richly generated audio experience. The novelty lies in integrating these aspects: prompt-based generative audio (Lyria), physiological confidence filtering, and a combination of rule-based safety with learned personalization (RL). We will provide evidence (via the literature on adaptive music and VR relaxation) that this multi-faceted approach should yield better engagement and outcomes than unadapted baselines[3][21].

2. OBJECTIVE

2.1 Main Objective

To design and implement a confidence-aware adaptive audio controller that generates and adjusts a personalized meditation audio stream in real time within a VR environment, using user preferences, stress level inputs, and signal confidence to optimize relaxation and engagement.

This objective is measurable in terms of system functionality (meeting design goals) and user study metrics (improved relaxation vs. baseline).

2.2 Specific Objectives

1. **User Preference Profiling:** Develop a mobile app module to collect user audio preferences (e.g. favorite instrument types, natural sounds, rhythm styles) and encode these into a structured audio profile or prompts for generative models.
2. **Base Audio Generation:** Implement integration with Google's Lyria RealTime API (or equivalent) to generate an initial personalized meditation soundtrack based on the user profile prompts.
3. **Physiological Input Handling:** Interface with Component B's stress level output and Component A's confidence score. Preprocess these signals (normalize, time-smooth).
4. **Rule-Based Adaptive Controller:** Design safe rule-based rules (e.g. limit rate of change) that adjust audio parameters (volume, intensity, brightness, ambient mix) in real time based on stress and confidence. Ensure no abrupt or extreme changes, especially when confidence is low.
5. **Reinforcement Learning Controller:** Formulate an RL problem (state = recent stress/confidence history, audio params; action = adjustments) and train an agent (possibly in simulation first) to learn a personalized adaptation policy that maximizes user comfort (feedback) while respecting safety constraints. Compare RL to rule-based.
6. **Novelty Management:** Implement a habituation-aware strategy that periodically reintroduces slight variations (e.g., regenerate a segment with similar prompt but different random seed) to maintain attentional interest without startling the user.
7. **Joint Audio Mixing:** Integrate control of the VR ambient audio (e.g. nature sounds, spatial cues) along with meditation music, adjusting their mix according to context (e.g., increase ambient subtlety if meditation layer intensifies).
8. **System Integration:** Embed the controller into Unity or Unreal. Use audio middleware (Wwise or FMOD) for low-latency parameter changes and spatialization. Ensure real-time performance (low latency <100ms) and security.
9. **Data Collection & Ethics:** Plan user studies collecting physiological data, preference/feedback, and subjective reports. Address privacy (anonymize data), obtain informed consent, and consider vulnerable user safeguards (e.g., content review).
10. **Evaluation:** Define metrics (e.g. relaxation score improvement, immersion rating, adaptation stability measures, latency, user satisfaction). Compare: static audio vs rule-based vs RL-based; high confidence vs low confidence input scenarios. Possibly use public datasets for baseline comparison of stress-adaptive audio (if available).

3. METHODOLOGY

Research Approach: We adopt an applied research and prototyping approach. This is primarily design and systems development (building the audio controller), supplemented by user-centered evaluation. It is interdisciplinary, combining AI audio generation, control systems, affective computing, and HCI.

System Design: We will follow a modular architecture. Key modules: (a) *Preference Manager* (collects and encodes user profile); (b) *Generative Music Engine* (Google Lyria RealTime API); (c) *Signal Interface* (ingests stress level & confidence scores); (d) *Adaptive Controller* (core of our component, with rule-based and RL submodules); (e) *Audio Mixer* (balances meditation music and ambient sounds via Wwise/FMOD in Unity).

Profile & Prompt Generation: The user preferences (from an app questionnaire) will include scales for desired audio intensity, preferred ambiance (e.g. nature vs synth), and familiarity vs novelty. These will be translated into text prompts or structured tags for Lyria. For example: "Ambient piano with light percussion, calm mood, medium tempo." We will experiment with prompt engineering techniques (iteratively refining phrasing) and possibly use a small local language model to optimize them[22]. The profile may also specify "avoid dissonance" for safety.

AI Sound Generation: Google Lyria RealTime will be used via its API or web interface in Unity. Lyria can generate continuous music with user controls[13][14]. We will configure it for the selected style (e.g., ambient nature sounds with soft instruments). The output stream will be captured and played through Wwise as an audio track. Preliminary testing ensures audio quality and timing meet requirements.

Rule-Based Control Design: We will define a set of rules based on domain knowledge and literature. For instance: if stress rises and confidence > threshold, gradually increase music intensity or add rhythmic elements; if stress drops or confidence is low, slowly fade to a simpler, calmer sound. Each audio dimension (intensity, density, brightness, volume balance) will have a safe change rate (Δ per second) to avoid jarring. These rule thresholds and gains will be tuned via pilot tests. Safety rules include limiting parameter jumps when confidence < e.g. 0.5, effectively "holding" state until reliable data. The controller will operate in real time (update at ~10Hz).

Reinforcement Learning Design: After rule-based testing, we will implement an RL agent to adapt parameters. State could include recent stress level, confidence, current audio parameter settings, and possibly time-in-session. Actions are discrete increments/decrements in key parameters. The reward function should encode user relaxation (for simulation, we might use a surrogate based on stress change; ultimately, user ratings). To bootstrap, we may use inverse RL or imitation of the rule-based policy, then refine it via user data. A safe-RL approach (constrained policy search) can ensure it doesn't violate comfort constraints[23]. Training could be in a simulated user model first, then gradually on actual users with off-policy updates.

Habituation/Novelty Strategy: We will track time elapsed on each musical segment. After a configurable duration (or number of bars), the system will generate a slight variation. For example, retrigger Lyria with the same prompt but a different seed or slight prompt tweak, or

alter one parameter (e.g., transpose key). This mimics the *dishabituation* effect[2][1], briefly shifting attention. Importantly, changes will be subtle (no drastic melody shifts) to maintain calm. If the user is highly relaxed for a long time (low stress and high confidence), we might deliberately introduce a small novel element to prevent mind-wandering (with the RL eventually learning the optimal timing and magnitude of such novelties).

Audio Mixing: The controller will adjust not just the generative music but also ambient sounds (e.g., forest, water, breath guide chimes). We will implement the audio mixer such that the "Meditation" track (Lyria) and "Environment" track have a blend ratio. For instance, during deep inhalation (low stress signal), the meditation track may dim slightly to allow ambient nature sounds to come forward, guiding breath. The parameters for ambient mix (volume, presence) will also adapt via the rule/RL system. Middleware like FMOD/Wwise can handle spatialization (e.g., 3D placement of ambient sources) for immersion.

Tools and Platforms: We will use Unity3D (or Unreal) for VR environment integration. Unity's audio pipeline plus an audio middleware (FMOD or Wwise) will provide low-latency parameter control and spatial mixing. Google Lyria RealTime will be accessed via its API or through Google's provided interfaces[13]. RL algorithms will be implemented in Python (TensorFlow/PyTorch) and interfaced via Unity ML-Agents or a custom plugin. The preference app can be a simple mobile/web UI (Flutter or web) for data collection.

Data Collection & Ethics: Users will be recruited with informed consent. Data collected includes audio preferences, stress/bio signals (from Component B/A), and subjective feedback (questionnaires on relaxation, presence, enjoyment). We will anonymize all data. Privacy is crucial: physiological data is sensitive, so it will be stored securely and used only for research. For safety, content will be non-triggering; also we will exclude individuals with severe epilepsy or similar conditions (VR warnings). An ethics review will be obtained.

Validation Strategy: We plan a mixed-methods evaluation. Objective metrics: reduction in stress (via validated scales or physiological proxies), consistency of adaptation (latency and stability of audio changes), and system performance (latency, CPU/GPU load). Subjective metrics: user-reported relaxation, immersion, and audio satisfaction on Likert scales. We will compare conditions in a within-subjects design: (1) Static audio (baseline meditation track), (2) Rule-based adaptive controller, (3) RL-enhanced adaptive controller. Each user will experience all conditions in counterbalanced order. We will also test scenarios with artificially lowered confidence to verify the controller's safe behavior. Statistical analysis (ANOVA) will assess significance. We may also compare to public datasets (if any) on music-adaptation (though none directly fit VR meditation).

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The system comprises **four main modules**: (1) **User Interface & Preference Module** (mobile app), (2) **Generative Music Engine** (Lyria), (3) **Adaptive Audio Controller** (our component), (4) **VR Integration & Mixing** (Unity with audio middleware).

Overall flow: The user first uses the mobile app to input audio preferences. These preferences are sent to our system and encoded into an initial *Audio Profile*. At run-time, the Generative Music Engine creates the base music layer. Meanwhile, Components A and B supply a real-time

Stress Level and a **Confidence Score** (0–1) at regular intervals. The Adaptive Audio Controller (AAC) receives these, along with the current audio parameters, and computes adjustments. It sends control commands to the Audio Mixer in Unity, which adjusts Lyria's output (e.g. changes instrument parameters or layering) and ambient tracks. The VR scene (lights, visuals) may also adjust modestly. The final stereo or spatial audio output goes to the user's HMD.

Component Diagram:
- *Mobile App:* Collects preferences (instrumental focus, mood, tempo preference, novelty tolerance) via survey. Outputs a structured profile (e.g. JSON).
- *Profile Engine:* Converts profile to Lyria prompts (using prompt engineering heuristics).
- *Lyria RealTime Engine:* Takes prompts, generates continuous audio. Exposes control parameters for intensity/density/brightness etc via API[13].
- *Signal Interface:* Receives from Component A (physio confidence) and B (stress index). Applies filters (e.g. moving average) to smooth the inputs.
- *Adaptive Controller:* Central module. Contains: - **Rule-Based Submodule:** A finite-state or fuzzy logic system encoding safe changes (e.g., "If stress ↑ and confidence >0.7, then raise intensity by 10% per second"). - **RL Submodule:** Policy network that learns to adjust parameters. Initially inactive, it learns from rewards (user feedback) and gradually overrides rule outputs in safe range. - **Novelty Manager:** Monitors duration; periodically signals to variation routines (e.g., modify prompt or effect). - *Audio Mixer:* Implemented with FMOD/Wwise. Receives parameter updates (volume, low-pass filter cutoff, reverb level). Also mixes in ambient sounds with controllable levels.
- *VR Environment:* Unity scene with visuals that may reflect stress (e.g. color hue, particle flow) and incorporate spatial audio.

Data Flow Example: Suppose user's stress increases. The signals arrive at AAC. If confidence is high, AAC might, via rules, increase music brightness and volume gradually to energize (or decrease, depending on design). If confidence is low, AAC would hold the current state (safety). The RL agent will note the context (e.g. previous successful adjustments for this user) to refine future actions. Meanwhile, if a segment has played over 2 minutes, Novelty Manager triggers a slight random variation.

Scalability & Security: The architecture is modular and event-driven, so it can scale (e.g. add more parameters or AI models). Sensitive data (physiology, prefs) is stored encrypted and used only in-session. The rule-based fallback ensures safety even if ML modules fail.

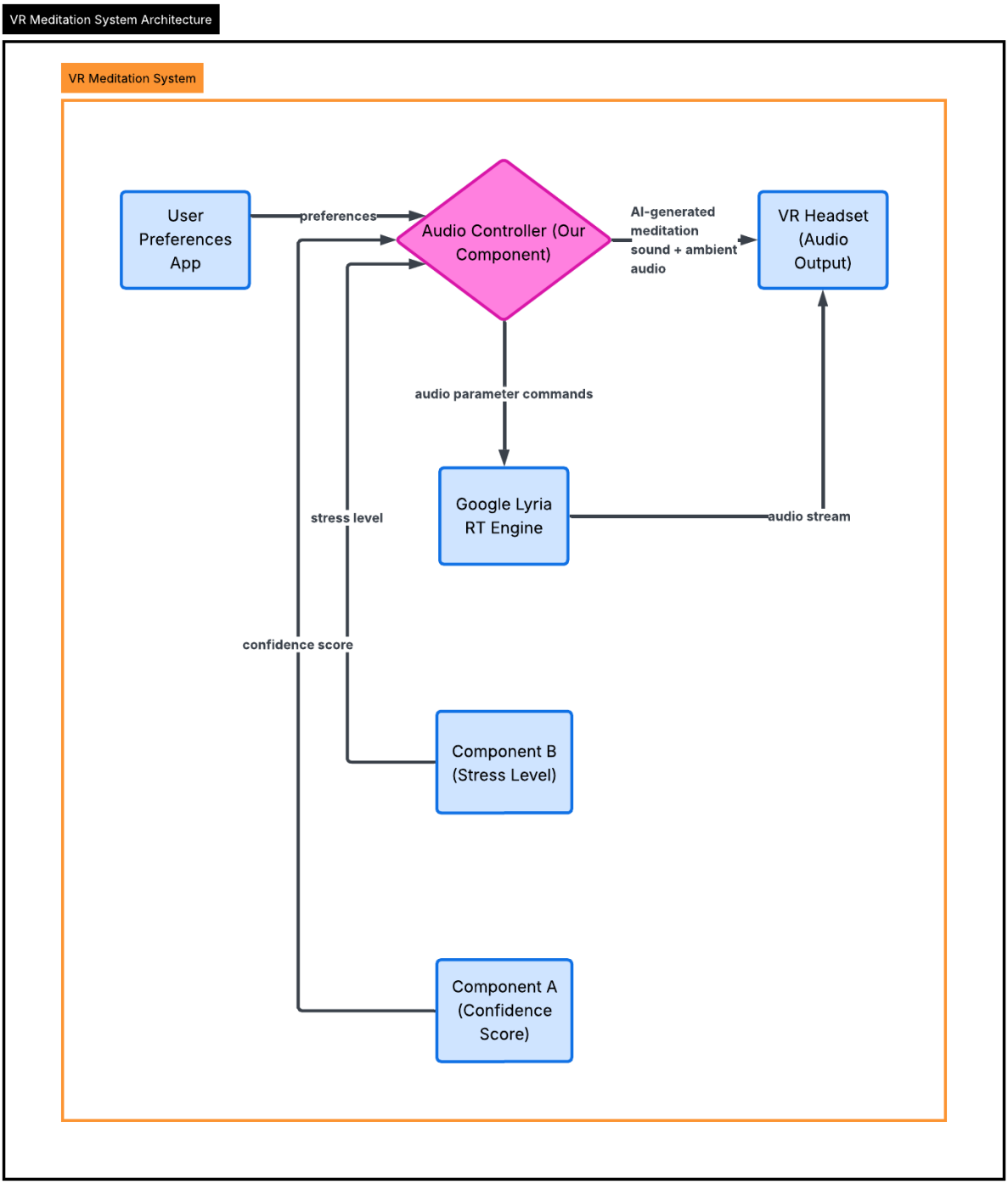


Figure 1- System-level architecture. The personalized audio controller (highlighted) interfaces with user preferences, physiological inputs, the AI audio engine, and VR audio output.

Figure 2- System-level architecture. The personalized audio controller (highlighted) interfaces with user preferences, physiological inputs, the AI audio engine, and VR audio output.

5. USER REQUIREMENTS

Stakeholders: End users (meditation practitioners), therapists, VR content developers, product managers. The user requirement elicitation would involve interviews and surveys with potential users (e.g. students, professionals with stress).

Functional Requirements (FR):
1. **FR1:** The system shall generate a personalized ambient music track based on user preferences.
2. **FR2:** The system shall receive and process real-time stress-level input from external Component B.
3. **FR3:** The system shall receive and use a physiological signal confidence score from Component A.
4. **FR4:** The system shall adapt audio parameters (intensity, brightness, density, volume, ambient mix) in real time based on stress and confidence.
5. **FR5:** The system shall enforce safety constraints (limiting speed/magnitude of changes) when confidence is low.
6. **FR6:** The system shall allow user override or mute of audio for safety/consent (non-emergency).
7. **FR7:** The system shall record session data (with consent) for analysis (timestamps, states, parameters).

Non-Functional Requirements (NFR):
1. **NFR1 (Usability):** The adaptation should feel natural and not distract. Users should be able to adjust preference settings easily (mobile UI).
2. **NFR2 (Latency):** Audio parameter changes should be applied within 100 ms of signal input to maintain real-time responsiveness.
3. **NFR3 (Privacy):** All personal data (physiological, preferences) shall be anonymized and encrypted in storage and transit.
4. **NFR4 (Reliability):** The system shall degrade gracefully: if Lyria fails, fallback to a stored audio loop. If input signals are lost, maintain last state.
5. **NFR5 (Safety):** The system shall never produce sudden loud or jarring audio shifts. It shall include an “emergency off” trigger in the VR environment.
6. **NFR6 (Adaptability):** The RL model should adapt over multiple sessions to an individual’s responses (learning rate, exploration policy).
7. **NFR7 (Security):** The mobile app and system interfaces should use secure authentication to protect user profiles.
8. **NFR8 (Regulatory):** The design should consider accessibility (e.g., alternative visual cues if audio changes are subtle), and compliance with medical device guidelines if marketed for health.

6. COMMERCIALIZATION PLAN

Market & Persona: Target markets include wellness app users, mental health clinics, and enterprise wellness programs. Customer persona: *"Stress-conscious professionals"* aged 25–45 who practice mindfulness and use tech for self-care. They value personalization and are willing to pay for premium mental health experiences.

Value Proposition: This component enhances VR meditation by delivering a unique, personalized audio experience that adapts safely to the user's current state. It differentiates from static VR apps by making each session custom to the individual's preferences and physiological needs.

Revenue Model: As part of a VR meditation platform, possible models are: subscription (monthly fee for continuous content updates and personalized training), or licensing to VR hardware vendors or wellness centers. An alternative is a one-time integration license fee plus maintenance.

Cost Estimation: Primary costs: development team (audio engineers, ML specialists), Google Lyria API usage (if priced per token or time), Unity licenses (if not free). Also hardware for testing (VR headsets, high-end PCs). These will be justified relative to market size; e.g., a subscription model at $10/month, targeting thousands of users yields significant potential.

Pricing Strategy: For direct-to-consumer, offer a basic free mode (static audio) and premium adaptive audio controller as a paid upgrade. For enterprise/clinical clients, custom deployment with license fee. The component's novelty (AI-personalized VR relaxation) and improved outcomes justify a premium price.

Competitive Advantage: Unlike competitors with fixed soundtracks, our system is AI-driven and personalized. Hybrid rule/RL ensures safe yet optimized adaptation, and the confidence-aware feature is unique, reducing risk of "bad data" causing harm. The component's IP could involve the adaptive algorithm and data representations (subject to patent review).

Intellectual Property: The specific hybrid control algorithm and user profiling method could be patentable. We will consult SLIIT's tech transfer office on IP strategy, possibly patenting the "confidence-aware audio control" concept if novel. Data (user preferences) is proprietary and protected.

(*The business plan would emphasize how this component enables selling the overall VR meditation system as an advanced solution.*)

7. BUDGET AND JUSTIFICATION

Budget Table (LKR)

Item	Qty	Unit Cost (LKR)	Total (LKR)	Justification
Development Laptop / PC (mid-range)	1	180,000	180,000	Mid-range laptop/desktop sufficient for Unity development, mobile app testing, and prototype-level AI/audio integration. Existing university lab resources can reduce heavy hardware dependence.
VR Headset (budget option / borrowed lab device)	1	165,000	165,000	Low-cost standalone VR headset or shared institutional device for prototype testing of immersive audio in VR.
Wwise Audio Middleware	1	0	0	Free/educational version can be used for undergraduate research purposes.
Unity License	1	0	0	Unity Personal edition is sufficient for student prototype development.
Android Smartphone (mid-range)	1	60,000	60,000	Required for testing the mobile preference input application and user interaction workflows.
iOS Device	1	85,000	85,000	Low-cost used/refurbished iPhone for basic compatibility testing, only if cross-platform validation is needed.
Audio Output Device (headphones/speakers)	1	25,000	25,000	Basic quality headphones/speakers for evaluating generated meditation audio and sound layering.
Cloud Server / Compute	3 months	10,000	30,000	Low-tier cloud compute or student credits for model testing, integration, and limited experimentation.
Data Storage / Hosting	3 months	2,000	6,000	Basic cloud storage/database/hosting for user preferences and prototype assets.
Generative Audio API / AI Service	3 months	15,000	45,000	Small prototype allocation for limited API experimentation. If unavailable, open-source or offline alternatives can be used.
Development Effort	1 lump sum	600,000	600,000	Estimated student/research development effort covering implementation, integration, debugging, and documentation.
Testing and Evaluation	1 lump sum	120,000	120,000	Covers user testing support, pilot study preparation, data collection, and analysis.

Justification

This project budget has been intentionally minimized to reflect the scope of an undergraduate research prototype rather than a full-scale commercial deployment. Expensive enterprise-grade tools and premium hardware have been replaced with student-appropriate alternatives such as mid-range computing devices, Unity Personal, educational/free middleware versions, limited cloud resources, and prototype-scale API usage. The budget assumes that some resources, including software access, testing space, and possible VR equipment support, may be partially obtained through university facilities or shared lab access. The largest allocation is assigned to development effort because system integration, immersive audio control logic, mobile preference handling, and evaluation activities represent the most time-intensive parts of the research.

8. WORK BREAKDOWN STRUCTURE (WBS)

1. **Preference and Profile Module**
   1.1. Design user preference survey UI
   1.2. Implement mobile app (Android/iOS)
   1.3. Profile encoding and prompt generation

2. **Audio Generation Setup**
   2.1. Integrate Google Lyria API in Unity
   2.2. Test base music generation with sample prompts
   2.3. Tune prompt-engineering for targeted mood/styles

3. **Signal Interface**
   3.1. Define stress level and confidence data formats
   3.2. Implement real-time signal receiver (simulate data initially)

4. **Rule-Based Controller**
   4.1. Define rule set and parameter limits (safety constraints)
   4.2. Implement rule logic for parameter adjustment
   4.3. Test and iterate rules with simulated data

5. **Reinforcement Learning Module**
   5.1. Define state, action, reward structure
   5.2. Develop simulation environment for training (simulate user responses)
   5.3. Train initial RL agent offline
   5.4. Integrate RL agent into Unity (ML-Agents or plugin)

6. **Novelty Management System**
   6.1. Design dishabituation triggers and magnitude
   6.2. Implement re-generation or parameter modulation routine
   6.3. Test impact on user attention with pilot users

7. **Audio Mixing and Integration**
   7.1. Set up FMOD/Wwise in Unity
   7.2. Connect audio tracks (meditation vs ambient)
   7.3. Implement spatialization for environment sounds
   7.4. Add controls for fade/mix ratio

8. **System Integration**
   8.1. Combine modules into single Unity project
   8.2. Ensure real-time performance (optimize code)
   8.3. Implement fail-safes and logging

9. **Data Collection & Ethics**
   9.1. Design user study protocol (IRB form, consent forms)
   9.2. Pilot recruitment and testing
   9.3. Data storage and analysis scripts

10. **Validation & Evaluation**
    10.1. Conduct user study (baseline, rule-based, RL conditions)
    10.2. Analyze quantitative metrics (relaxation scores, adaptation latencies)
    10.3. Collect qualitative feedback (interviews, surveys)

11. **Report Writing**
    11.1. Draft dissertation proposal report sections
    11.2. Revise with supervisor feedback
    11.3. Final proofreading and formatting

Milestones: Prototype of adaptive controller (end Month 3), RL integration (Month 5), user study completion (Month 6), final report draft (Month 7).

9. Adaptive Audio Controller – Component Breakdown

- **Purpose:** Detail internal submodules of our component: rule-based controller, RL agent, novelty manager, audio mixer interface.
- **Audience:** Technical reviewers/architects.
- **Detail:** *Detailed.* Show data/control flow among submodules. Inputs: user profile, stress, confidence. Outputs: final audio parameters.
- **Elements:**
- **User Profile Handler** (preference parsing, prompt generation).
- **Rule-Based Engine** (safe adaptation logic).
- **RL Adaptation Agent** (learning policy).
- **Novelty Manager** (habituation handling).
- **Audio Parameters Output** block.
- Edges:
    - Profile input -> Rule & RL (initial prompts).
    - Stress, Confidence -> both Rule and RL.
    - Rule & RL output -> merging / arbitration -> Audio Params.
    - Novelty manager taps into Audio Params and modifies if needed.

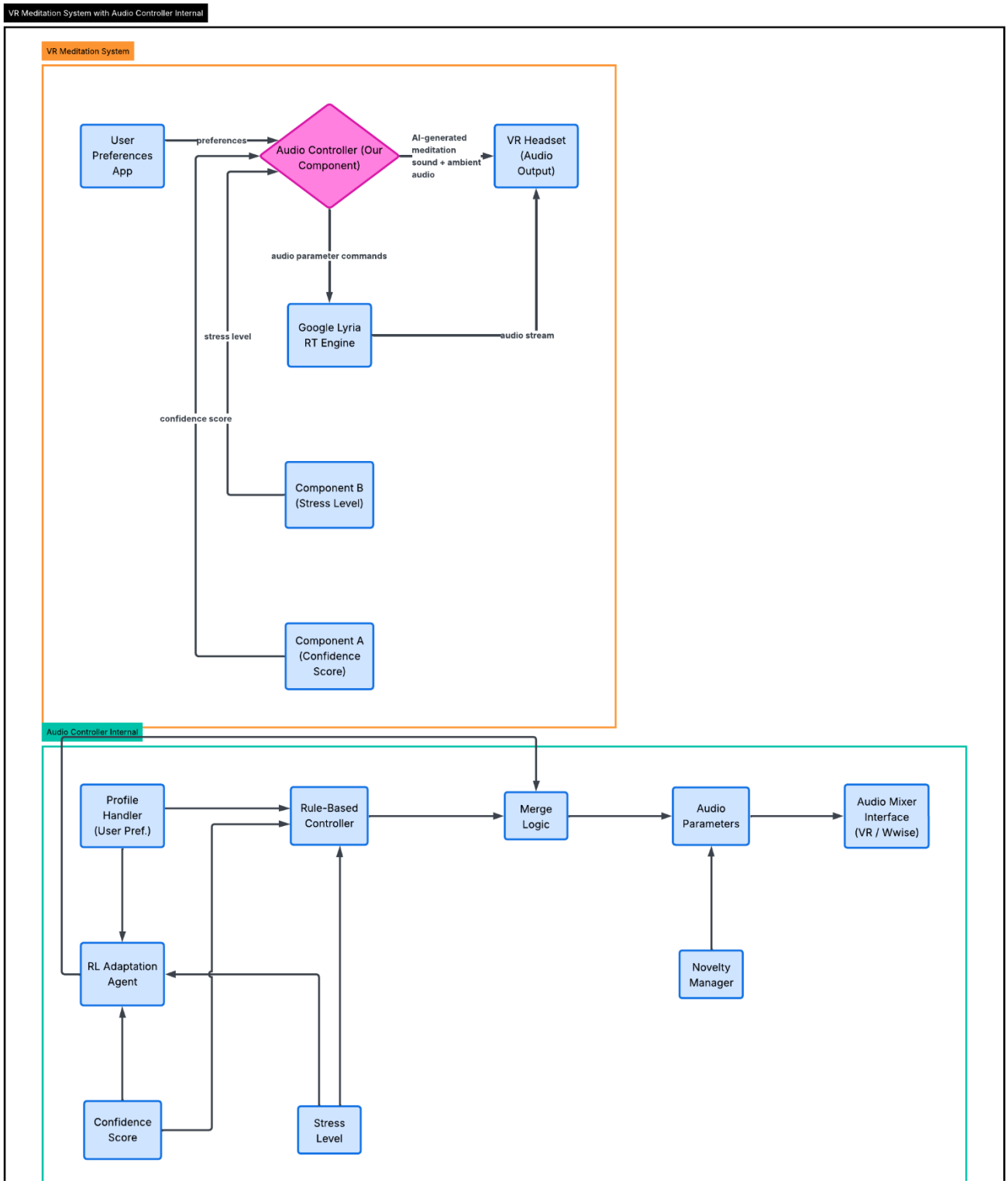


Figure 3- Internal component diagram of the Adaptive Audio Controller. User profile and sensor inputs feed both rule-based and RL submodules, whose outputs are merged into final audio parameters, with a novelty manager adjusting parameters over time.

Figure 4- Internal component diagram of the Adaptive Audio Controller. User profile and sensor inputs feed both rule-based and RL submodules, whose outputs are merged into final audio parameters, with a novelty manager adjusting parameters over time.

10. Audio Adaptation Sequence (Interaction Flow)

- **Purpose:** Show time-sequenced interactions among key parts during a meditation session.
- **Audience:** Technical evaluators verifying dynamic behavior.
- **Detail:** *Medium.* Sequence from user → system: preferences upload, real-time sensor updates, audio adjustments, and output.
- **Participants:** UserApp, Component B (Sensors), Component A (Sensors), AudioController, Lyria Engine, VR/Audio Output.

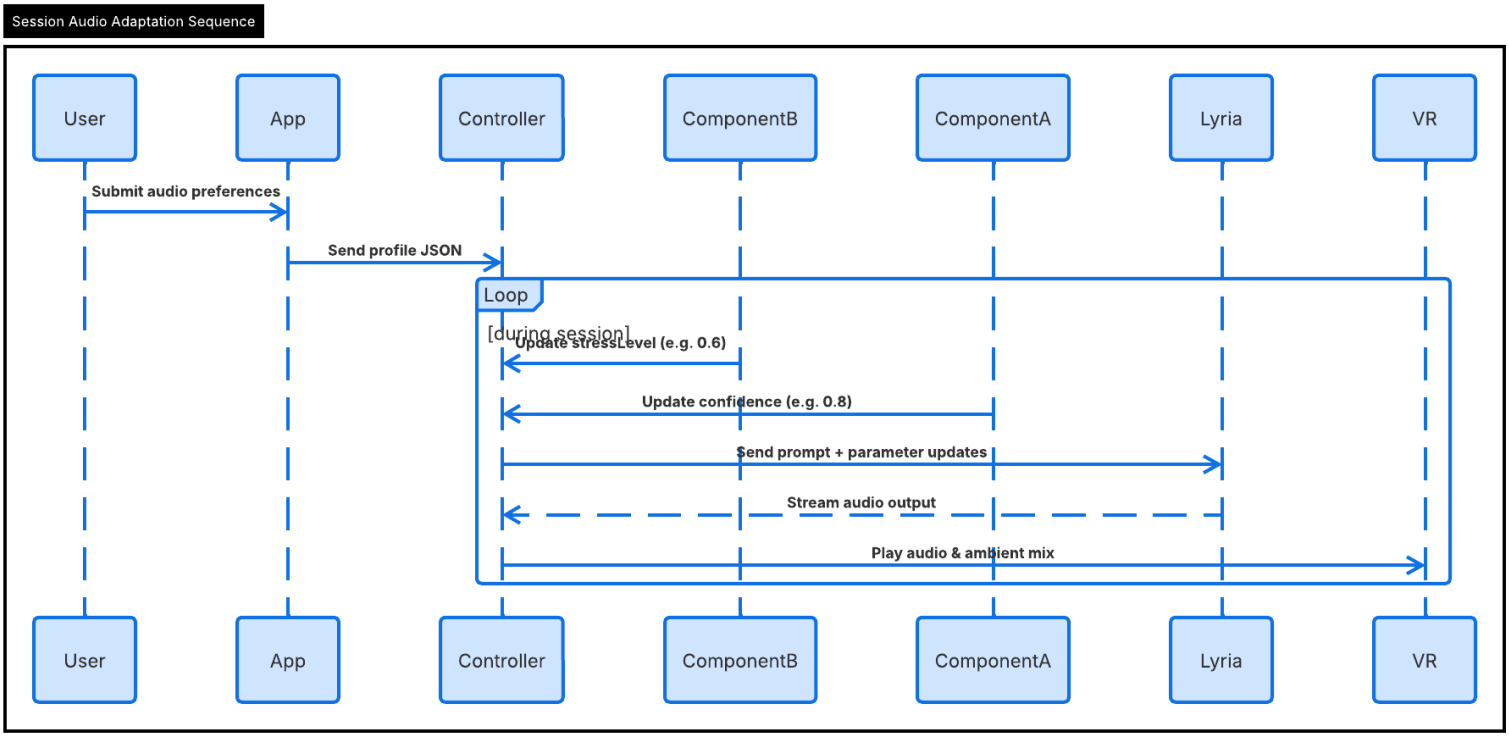


Figure 5 - Sequence diagram of runtime adaptation. The system receives user preferences and continuous sensor updates, processes them in the controller, adjusts the AI-generated audio via Lyria, and outputs sound to the VR headset.

11. GANTT CHART

The table below outlines the timeline (months) for tasks (1-11 from WBS). Each row spans planned activity duration.

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Milestone
1. Preference Module	■■■							Preferences App Ready
2. Audio Generation Setup	■■■	■■■						Base Audio Generation
3. Signal Interface		■■						Signals Receiving
4. Rule-Based Controller		■■■	■■■					Rule Controller Ready
5. Reinforcement Learning Module			■■	■■■	■■■			RL Agent Integrated
6. Novelty Management			■■	■■				Novelty System Active
7. Audio Mixing Integration	■■■	■■						Mixer Configured
8. System Integration		■■	■■■	■■				System Prototype v1.0
9. Data Collection & Ethics			■■	■■	■■			Ethics Approval + Pilot
10. Validation & Evaluation					■■	■■■		User Study Completed
11. Report Writing						■■	■■	Final Report Submission

(Legend: ■ indicates active work. Milestones noted in last column.)

12. REFERENCES

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(Additional references on meditation and audio techniques are cited in context.)

13. APPENDICES

A. Sample User Preference Questionnaire: (Example questions on preferred music genres, instruments, tempo, relaxation goals)

B. Adaptive Audio Parameter Table: (E.g., mapping stress levels to music intensity and ambient volume)

C. System Overview Diagram Description: (Description of Figure 1; e.g. “The figure shows the Audio Controller receiving inputs (prefs, stress, confidence) and outputting audio parameter controls to the VR engine.”)

D. Risk Analysis Table: (E.g., risk of over-reliance on noisy signals – mitigated by confidence threshold rules.)

E. Ethical Considerations Checklist: (Consent procedures, data privacy, safety protocols for vulnerable users.)

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